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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

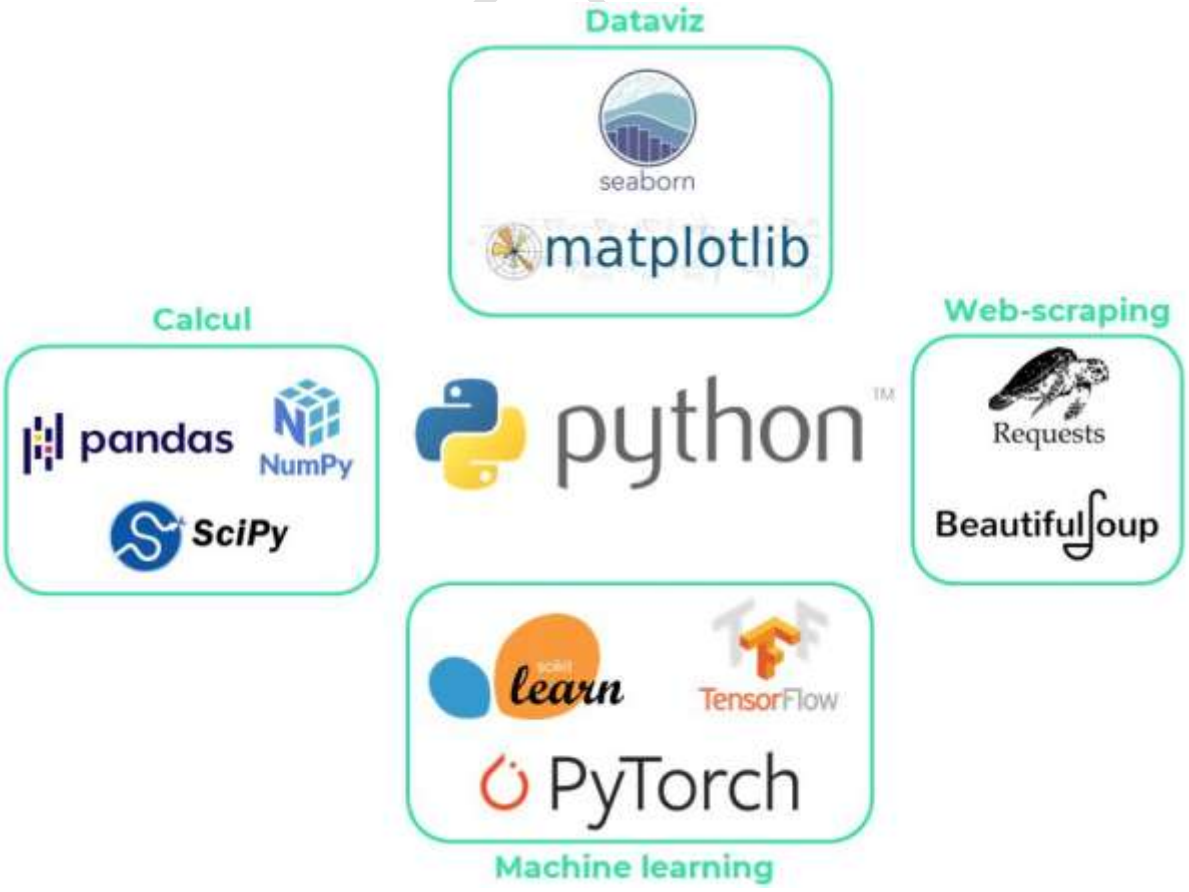
Classes : Offline & Online

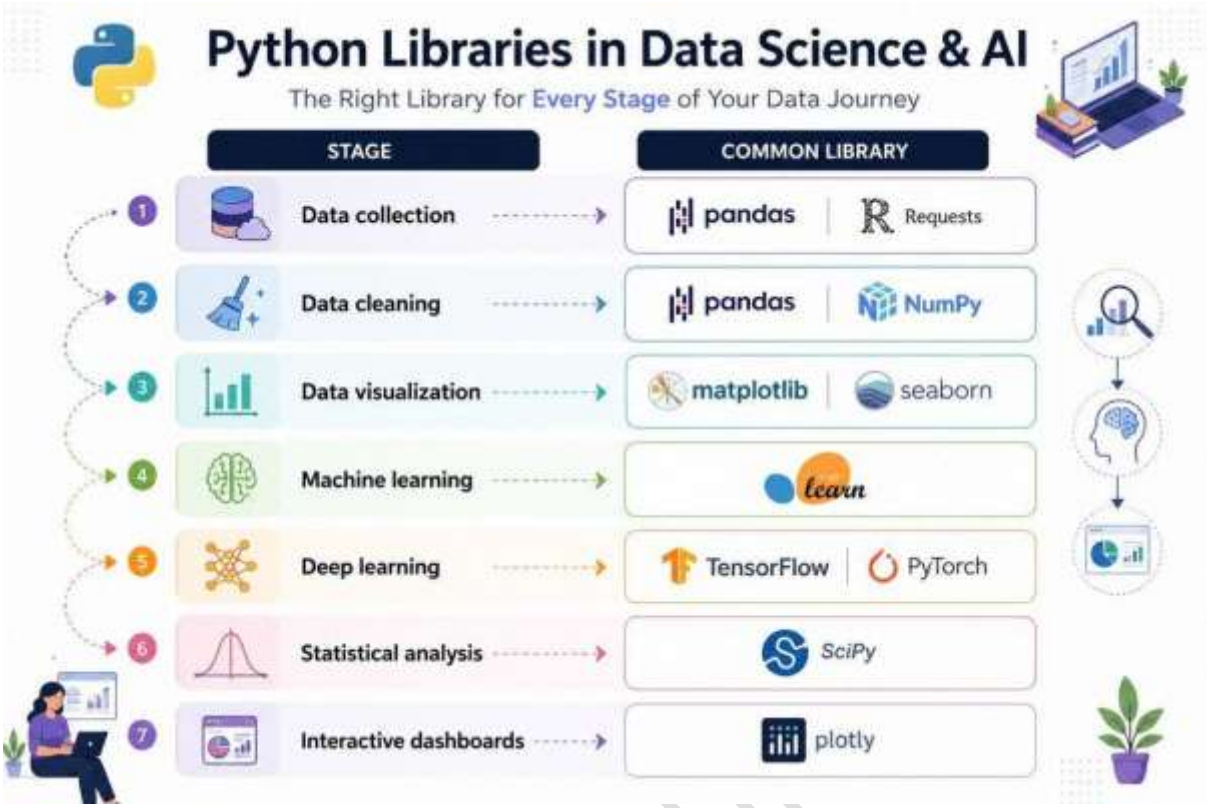
Daily Teaching : 2 Hours

Total Classes : 100

Level 4 – AI Professional Foundation(11th–12th Standard)

Class 23 – Python Libraries





1. Learning Objectives

By the end of this lesson, students will be able to:

- Understand what a **Python library** is.
- Understand the difference between modules, packages, and libraries.
- Import and use Python libraries.
- Explore useful built-in libraries.
- Use math, random, datetime, and os.
- Understand popular third-party libraries.
- Use NumPy for numerical operations.
- Use Pandas for data processing.
- Use Matplotlib for visualization.

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- Understand where libraries are used in real-world applications.
- Create a small **Student Data Analysis Project**.

2. What is a Python Library?

A **Python library** is a collection of ready-made code that programmers can use to perform specific tasks.

Instead of writing everything from the beginning, we can use existing functionality.

Simple example

Suppose we want to calculate the square root of 144.

Without a library, we would need to write our own calculation.

With Python's math library:

```
import math


print(math.sqrt(144))
```

Output

```
12.0
```

The library provides ready-made functionality.

3. Why Do We Use Libraries?

Libraries help programmers:

- Save time
- Reduce code

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- Solve complex problems
- Reuse existing code
- Build applications faster
- Perform calculations
- Process data
- Create charts
- Work with files
- Connect to the internet
- Build AI and ML applications

Basic idea

Without Library

↓

Write everything yourself

↓

More code + More time

With Library

↓

Use ready-made functionality

↓

Less code + Less time

4. Real-World Example

Imagine you are building a school management application.

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You may need:

Math calculations → math

Random student ID → random

Date of admission → datetime

Files/folders → os

Student datasets → pandas

Charts → matplotlib

Numerical calculations → numpy

Instead of building these features from scratch, libraries provide useful tools.

5. Module vs Package vs Library

Students often confuse these terms.

Term	Meaning	Example
Module	Python file containing code	math
Package	Collection of related modules	numpy
Library	Collection of reusable functionality	NumPy, Pandas
Framework	Structure for building applications	Django

For school-level understanding:

A library is a collection of reusable code that helps us perform specific tasks.

6. How to Import a Library

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The basic syntax is:

import library_name

Example:

import math

Then use:

math.sqrt(25)

7. Import Specific Functions

We can import a particular function.

from math import sqrt

print(sqrt(25))

Output

5.0

Instead of:

math.sqrt(25)

we can directly use:

sqrt(25)

8. Import with an Alias

An alias gives a shorter name.

import math as m

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print(m.sqrt(100))

Output

10.0

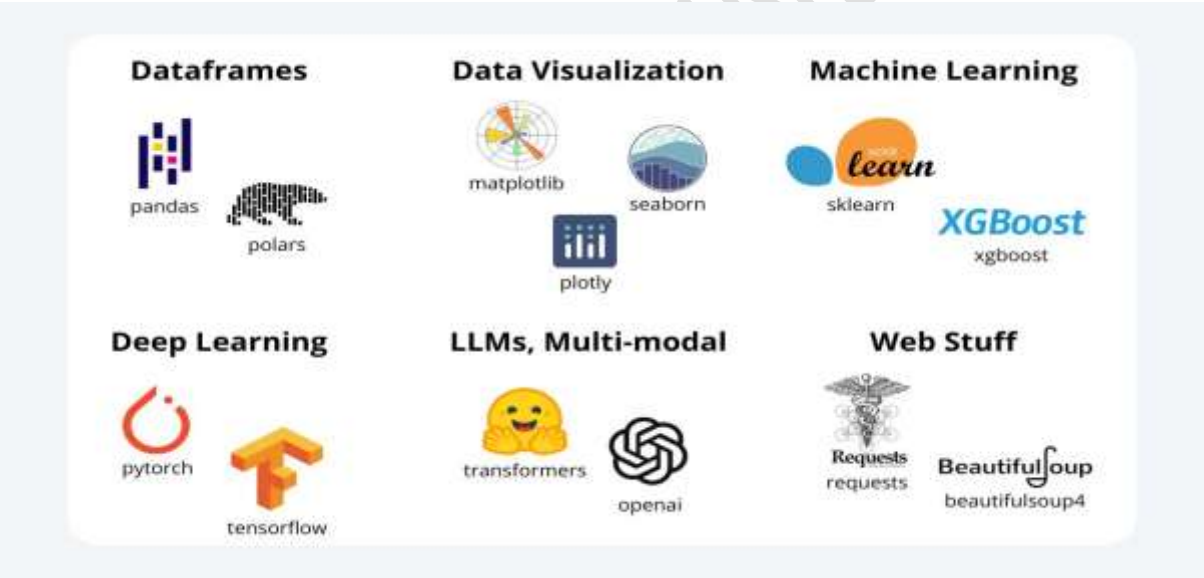
A very common example is:

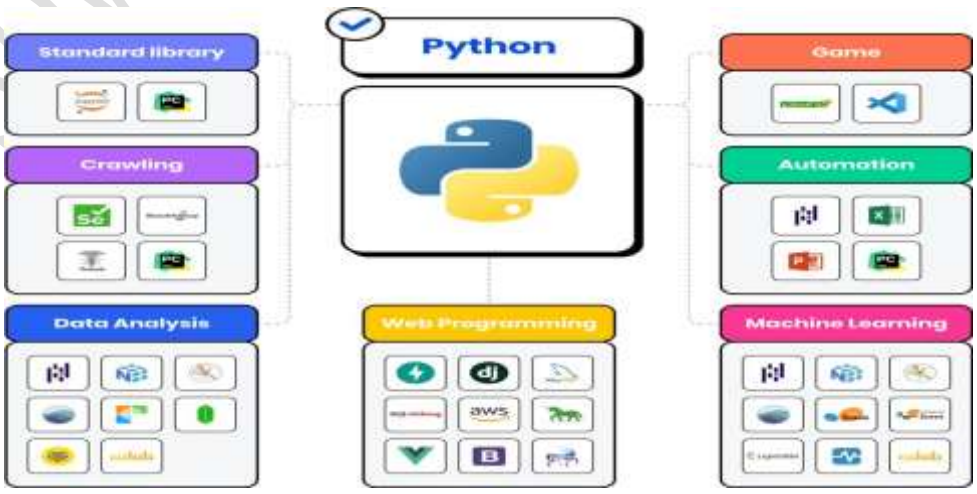
import numpy as np

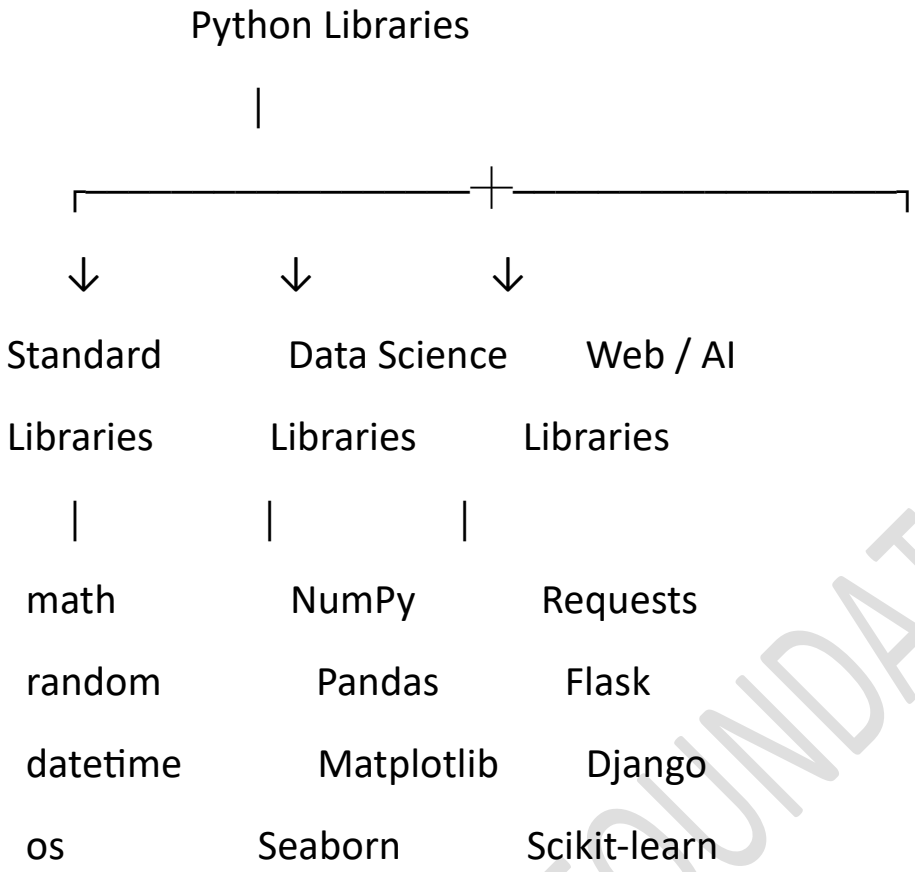
and:

import pandas as pd

9. Python Library Categories







10. Useful Python Libraries

Here are some important libraries for 11th–12th standard students:

Library	**Main Use**
math	Mathematical operations
random	Random values
datetime	Date and time
os	Operating system/file operations
json	JSON data
csv	CSV files

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Library	Main Use
statistics	Statistical calculations
numpy	Numerical computing
pandas	Data analysis
matplotlib	Data visualization
requests	Internet/API requests
scikit-learn	Machine Learning

11. Library 1 – math

The math module provides mathematical functions.

```
import math


print(math.sqrt(81))
print(math.pow(2, 3))
print(math.factorial(5))
print(math.pi)
```

Output

```
9.0
8.0
120
3.141592653589793
```

Useful Functions

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Function	Purpose
sqrt()	Square root
pow()	Power
factorial()	Factorial
ceil()	Round upward
floor()	Round downward
pi	Value of π

12. Practical Activity – Math Library

```
import math
number = float(input("Enter a number: "))
print("Square Root:", math.sqrt(number))
print("Square:", math.pow(number, 2))
print("Ceiling:", math.ceil(number))
print("Floor:", math.floor(number))
```

Sample Output

```
Enter a number: 7.5
Square Root: 2.7386127875258306
Square: 56.25
Ceiling: 8
Floor: 7
```

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13. Library 2 – random

The random module generates pseudo-random values.

Example:

```
import random

number = random.randint(1, 10)

print("Random Number:", number)
```

Possible output:

Random Number: 7

The result may be different each time.

14. Random Choice

```
import random

students = ["Asha", "Rahul", "Ravi"]

winner = random.choice(students)


print("Selected Student:", winner)
```

Possible output:

Selected Student: Ravi

15. Practical Activity – Dice Simulator

```
import random

dice = random.randint(1, 6)

print("You rolled:", dice)
```

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Possible output:

You rolled: 4

Mini Challenge

Create a program that simulates **two dice**.

```
import random

dice1 = random.randint(1, 6)

dice2 = random.randint(1, 6)


print("Dice 1:", dice1)

print("Dice 2:", dice2)

print("Total:", dice1 + dice2)
```

16. Library 3 – datetime

The datetime module is used to work with dates and times.

```
from datetime import datetime

now = datetime.now()

print("Current Date and Time:", now)
```

Possible output:

Current Date and Time: 2026-09-04 10:20:15

17. Display Date Separately

```
from datetime import datetime
```

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```
today = datetime.now()
print("Year:", today.year)
print("Month:", today.month)
print("Day:", today.day)
```

18. Student Admission Example

```
from datetime import datetime
student_name = input("Enter student name: ")
admission_date = datetime.now()
print("Student:", student_name)
print("Admission Date:", admission_date.date())
```

This is useful in:

- School management
- College management
- Employee management
- Hospital applications
- Booking systems

19. Library 4 – os

The os module allows Python to interact with the operating system.

For example:

```
import os
```

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```
print(os.getcwd())
```

This displays the current working directory.

Create a folder

```
import os

os.mkdir("StudentData")

print("Folder created")
```

If the folder already exists, this command will raise an error.

20. Check Whether a File Exists

```
import os


if os.path.exists("students.txt"):
    print("File exists")
else:
    print("File does not exist")
```

This is very useful in file-handling applications.

21. Library 5 – json

We studied JSON in an earlier class.

Python provides the json module for working with JSON data.

```
import json


student = {
    "name": "Asha",
```

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```
    "roll_no": 101,
    "marks": 90
}

data = json.dumps(student)

print(data)
```

Output

```
{"name": "Asha", "roll_no": 101, "marks": 90}
```

22. Library 6 – statistics

The statistics module provides common statistical calculations.

```
import statistics
marks = [80, 75, 90, 85, 70]

print("Mean:", statistics.mean(marks))
print("Median:", statistics.median(marks))
```

Output

```
Mean: 80
Median: 80
```

23. Third-Party Libraries

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Some libraries are not included in Python's basic installation and are commonly installed separately.

Examples:

NumPy

Pandas

Matplotlib

Requests

Scikit-learn

These are extremely useful in:

- Data Science
- Machine Learning
- Artificial Intelligence
- Data Visualization
- Web/API programming

24. Installing a Third-Party Library

We can commonly use pip.

Example:

pip install numpy

For Pandas:

pip install pandas

For Matplotlib:

pip install matplotlib

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What is pip?

pip is Python's package installer.

pip

↓

Downloads/installs

↓

Python package

↓

Use package in program

25. Library 7 – NumPy

NumPy stands for **Numerical Python**.

It is widely used for numerical and scientific computing.

Import:

import numpy as np

Create an array:

import numpy as np

marks = np.array([80, 75, 90, 85, 70])

print(marks)

Output

[80 75 90 85 70]

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26. NumPy Calculations

```
import numpy as np
marks = np.array([80, 75, 90, 85, 70])
print("Total:", np.sum(marks))
print("Average:", np.mean(marks))
print("Maximum:", np.max(marks))
print("Minimum:", np.min(marks))
```

Output

Total: 400

Average: 80.0

Maximum: 90

Minimum: 70

27. Why NumPy?

Without NumPy:

Large numerical data

↓

More manual processing

With NumPy:

Large numerical data

↓

NumPy arrays

↓

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Fast numerical operations

NumPy is especially important for future:

- Machine Learning
- Data Science
- Scientific Computing

28. Library 8 – Pandas

Pandas is widely used for data analysis and data processing.

Import:

```
import pandas as pd
```

Create a DataFrame:

```
import pandas as pd
data = {
   "Name": ["Asha", "Rahul", "Ravi"],
   "Marks": [90, 85, 78]
}
df = pd.DataFrame(data)
print(df)
```

Output

```
   Name  Marks
0   Asha     90
1  Rahul     85
2   Ravi     78
```

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29. Pandas DataFrame

A **DataFrame** looks similar to a table.

```
┌─────┬─────────┬─────────┐
│     | Name    | Marks   |
├─────┼─────────┼─────────┤
| 0   | Asha    | 90      |
| 1   | Rahul   | 85      |
| 2   | Ravi    | 78      |
└─────┴─────────┴─────────┘
```

It is useful for:

- Student records
- Employee records
- Sales data
- Attendance
- Survey data
- Business reports

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30. Pandas Data Analysis

```
import pandas as pd


data = {
   "Name": ["Asha", "Rahul", "Ravi", "Sara"],
```

```
    "Marks": [90, 85, 78, 95]
}
df = pd.DataFrame(data)
print("Average:", df["Marks"].mean())
print("Highest:", df["Marks"].max())
print("Lowest:", df["Marks"].min())
```

Output

```
Average: 87.0
Highest: 95
Lowest: 78
```

31. Filtering Data with Pandas

We can select students whose marks are greater than 80.

```
import pandas as pd
data = {
    "Name": ["Asha", "Rahul", "Ravi", "Sara"],
    "Marks": [90, 85, 78, 95]
}
df = pd.DataFrame(data)
top_students = df[df["Marks"] > 80]
print(top_students)
```

Output

```
    Name  Marks
```

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0 Asha 90

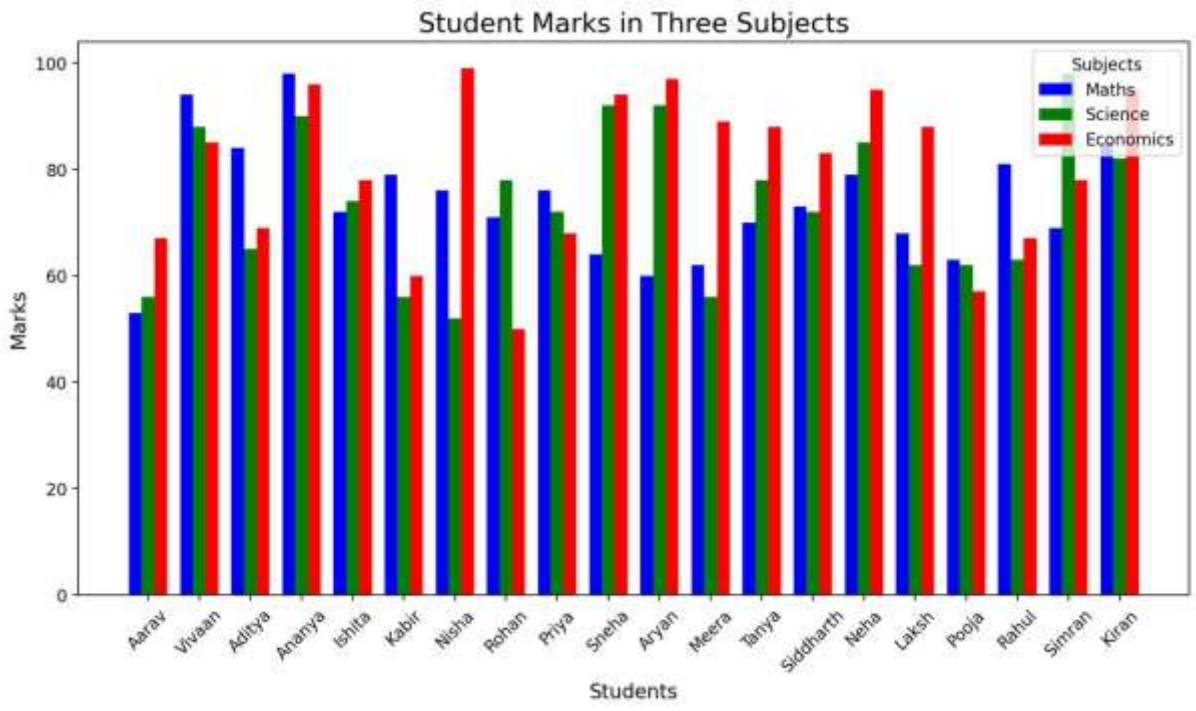
1 Rahul 85

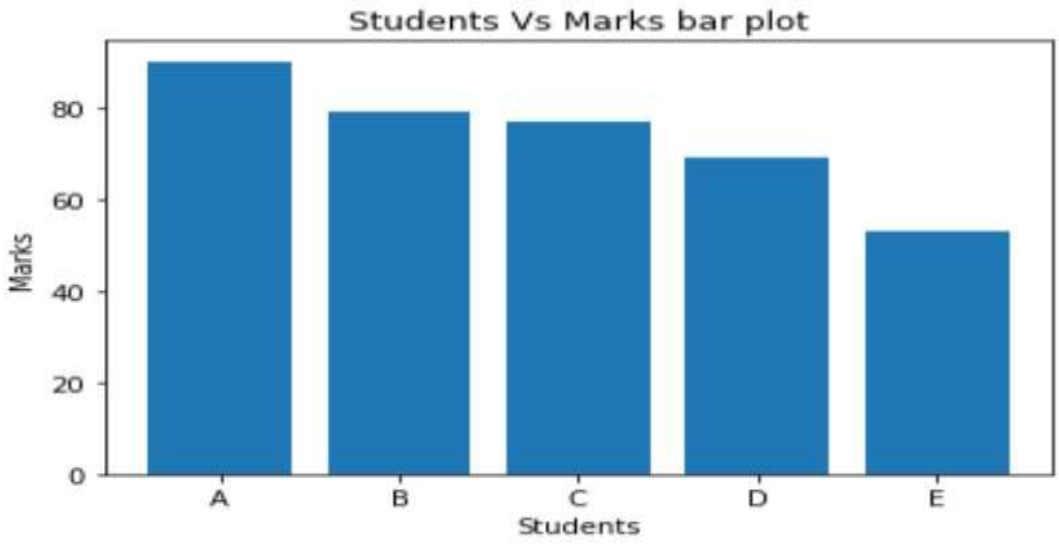
3 Sara 95

This is an example of **data processing**.

32. Library 9 – Matplotlib

Matplotlib is used to create graphs and charts.





Example:

```
import matplotlib.pyplot as plt


students = ["Asha", "Rahul", "Ravi", "Sara"]
marks = [90, 85, 78, 95]
plt.bar(students, marks)
plt.xlabel("Students")
plt.ylabel("Marks")
plt.title("Student Marks")
plt.show()
```

This creates a **bar chart** showing student marks.

33. Types of Charts

Matplotlib can create:

Bar Chart

Line Chart

Pie Chart

Scatter Plot

Histogram

For example:

```
plt.bar()
plt.plot()
plt.pie()
```

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plt.scatter()

plt.hist()

34. Library 10 – Requests

The requests library is commonly used to communicate with web servers and APIs.

Example:

```
import requests

response = requests.get("https://example.com")

print(response.status_code)
```

A successful response commonly returns:

200

Real-world applications

Requests can be used to retrieve information from:

- Weather APIs
- News APIs
- Public data APIs
- Web services

35. Library 11 – Scikit-learn

Scikit-learn is a popular machine-learning library.

It provides tools for:

- Classification

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- Regression
- Clustering
- Data preprocessing
- Model evaluation

Example concept:

Student Data

↓

Training Data

↓

Machine Learning Algorithm

↓

Model

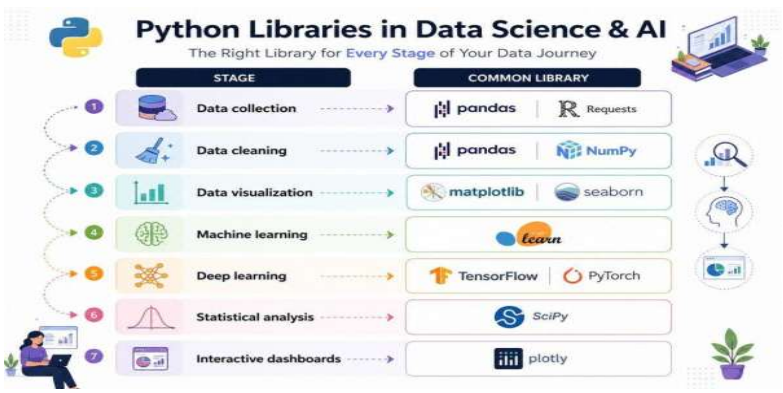
↓

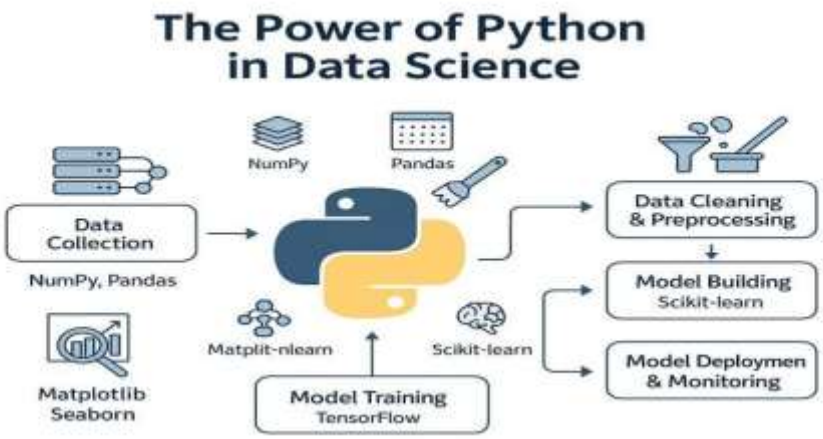
Prediction

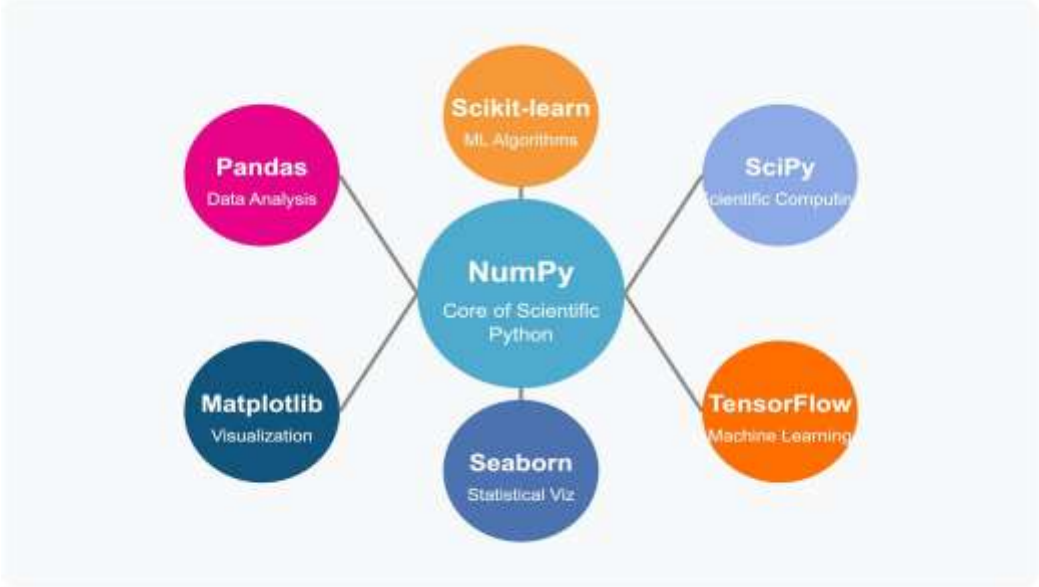
You will explore this more deeply in the **Machine Learning & AI module**.

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36. Library Selection Guide







What problem are you solving?



↓ ↓ ↓

Math Data Charts

↓ ↓ ↓

math Pandas Matplotlib

NumPy

|

↓

Machine Learning

↓

Scikit-learn

37. Complete Practical – Student Data Analysis

This project combines:

- Pandas
- NumPy
- Matplotlib

Program

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt


data = {
   "Name": ["Asha", "Rahul", "Ravi", "Sara", "John"],
   "Marks": [90, 85, 78, 95, 88]
}
df = pd.DataFrame(data)
marks = np.array(df["Marks"])
print("Student Data:")
```

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```
print(df)
print("\nAverage Marks:", np.mean(marks))
print("Highest Marks:", np.max(marks))
print("Lowest Marks:", np.min(marks))
plt.bar(df["Name"], df["Marks"])
plt.xlabel("Student")
plt.ylabel("Marks")
plt.title("Student Performance")
plt.show()
```

Program Flow

Student Data

↓

Pandas DataFrame

↓

NumPy Array

↓

Calculate:

├── Average

├── Maximum

└── Minimum

↓

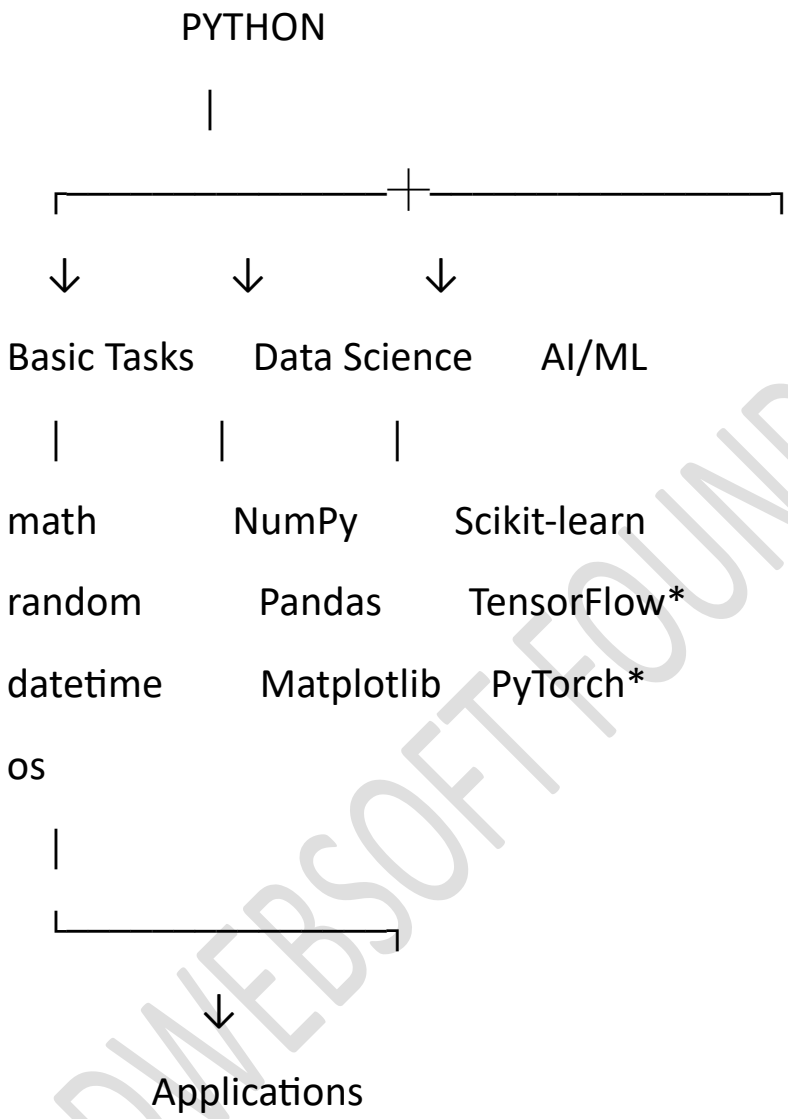
Matplotlib

↓

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Performance Chart

38. Real-World Library Ecosystem



*These are examples of libraries students may encounter later in advanced AI/ML learning.

Practical Session

39. Practical Activity 1 – Random Student Selection

Create a list:

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students = ["Asha", "Rahul", "Ravi", "Sara", "John"]

Use the random library to select one student.

Challenge

Modify the program to select **three students** without repeating names.

40. Practical Activity 2 – Student Statistics

Use:

import statistics

Given:

marks = [85, 90, 76, 88, 95]

Find:

- Mean
- Median
- Highest
- Lowest

41. Practical Activity 3 – Student DataFrame

Create a Pandas DataFrame with:

Name

Roll Number

Marks

Grade

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Example:

```
import pandas as pd

data = {
   "Name": ["Asha", "Rahul", "Ravi"],
   "Roll": [101, 102, 103],
   "Marks": [90, 85, 78],
   "Grade": ["A+", "A", "B"]
}

df = pd.DataFrame(data)

print(df)
```

42. Practical Activity 4 – Create a Chart

Using Matplotlib:

```
import matplotlib.pyplot as plt
subjects = ["Python", "Maths", "Science", "English"]
marks = [90, 85, 88, 92]
plt.bar(subjects, marks)
plt.title("Subject Marks")
plt.xlabel("Subjects")
plt.ylabel("Marks")
```

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plt.show()

Challenge

Create a chart for **five students** instead of subjects.

43. Common Mistakes

Mistake 1 – Forgetting to import

❌

print(math.sqrt(25))

if math hasn't been imported.

✅

import math

print(math.sqrt(25))

Mistake 2 – Wrong alias

If we write:

import pandas as pd

we should use:

pd.DataFrame()

not:

pandas.DataFrame()

unless pandas was also imported under that name.

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Mistake 3 – Package not installed

If Python shows something like:

ModuleNotFoundError

the required third-party library may not be installed in the current environment.

For example:

pip install pandas

Mistake 4 – Confusing a library with a function

For example:

math

is the module, while:

math.sqrt()

is a function from that module.

44. Standard Library vs Third-Party Library

Standard Library	Third-Party Library
Comes with Python	Usually installed separately
No separate installation normally needed	Often installed using pip
math	numpy
random	pandas

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Standard Library	Third-Party Library
datetime	matplotlib
os	requests
json	scikit-learn

45. Library Safety and Good Practices

Students should learn to use libraries responsibly.

1. Install trusted packages

Use official Python package sources and trusted project documentation.

2. Understand what you install

Don't install unknown packages just because someone sends a command.

3. Keep libraries updated

Updates can contain:

- Bug fixes
- Security fixes
- Improvements

4. Import only what you need

Clear imports make programs easier to understand.

46. Mini Project – Student Performance Analyzer

Objective

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Create an application that analyzes student marks.

Input Data

Name

Python

Maths

Science

English

Use

Pandas → Store/process data

NumPy → Numerical calculations

Matplotlib → Create charts

Required Features

1. Display all students.
2. Calculate average marks.
3. Find highest marks.
4. Find lowest marks.
5. Identify students above average.
6. Display a bar chart.

Project Flow

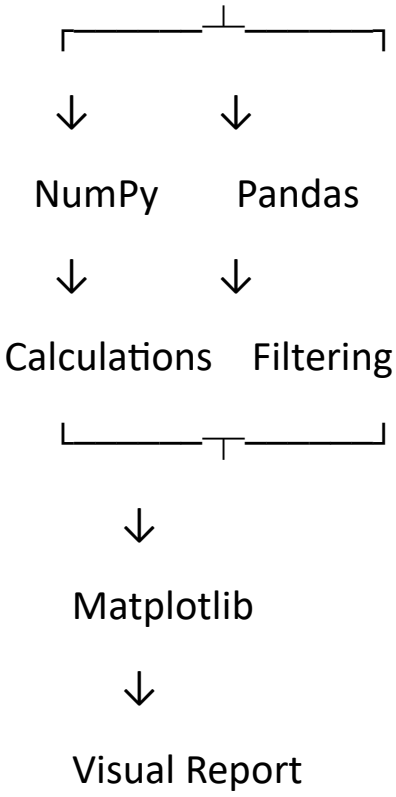
Student Data

↓

Pandas DataFrame

↓

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47. Quick Revision

What is a library?

A collection of reusable code that helps solve programming problems.

Import

import math

Specific import

from math import sqrt

Alias

import numpy as np

Important libraries

math → Mathematics

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random → Random values

datetime → Date/time

os → Operating system

json → JSON data

statistics → Statistics

NumPy → Numerical computing

Pandas → Data analysis

Matplotlib → Visualization

Requests → HTTP/API requests

Scikit-learn → Machine Learning

48. MCQs

1. What is a Python library?

A. A computer game
B. A collection of reusable code
C. An operating system
D. A database

Answer: B

2. Which library is used for mathematical functions?

A. math
B. pandas
C. random
D. os

Answer: A

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3. Which library is commonly imported as np?

A. Pandas
B. NumPy
C. Matplotlib
D. Requests

Answer: B

4. Which library is commonly imported as pd?

A. Pandas
B. NumPy
C. Math
D. OS

Answer: A

5. Which library is used for data visualization?

A. os
B. random
C. matplotlib
D. json

Answer: C

6. What is pip?

A. Python's package installer
B. A Python variable
C. A loop
D. A data type

Answer: A

7. Which library is commonly used for DataFrames?

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A. NumPy
B. Pandas
C. Math
D. Random

Answer: B

8. Which library is associated with machine learning?

A. Scikit-learn
B. os
C. datetime
D. json

Answer: A

49. Practical Assignment

Python Library Explorer

Create a Python program that demonstrates at least **five libraries/modules**.

Students should use:

1. math
2. random
3. datetime
4. statistics
5. pandas

Program should:

- Perform a mathematical calculation.

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- Generate a random number.
- Display today's date.
- Calculate average marks.
- Create a Pandas DataFrame.

Advanced Challenge

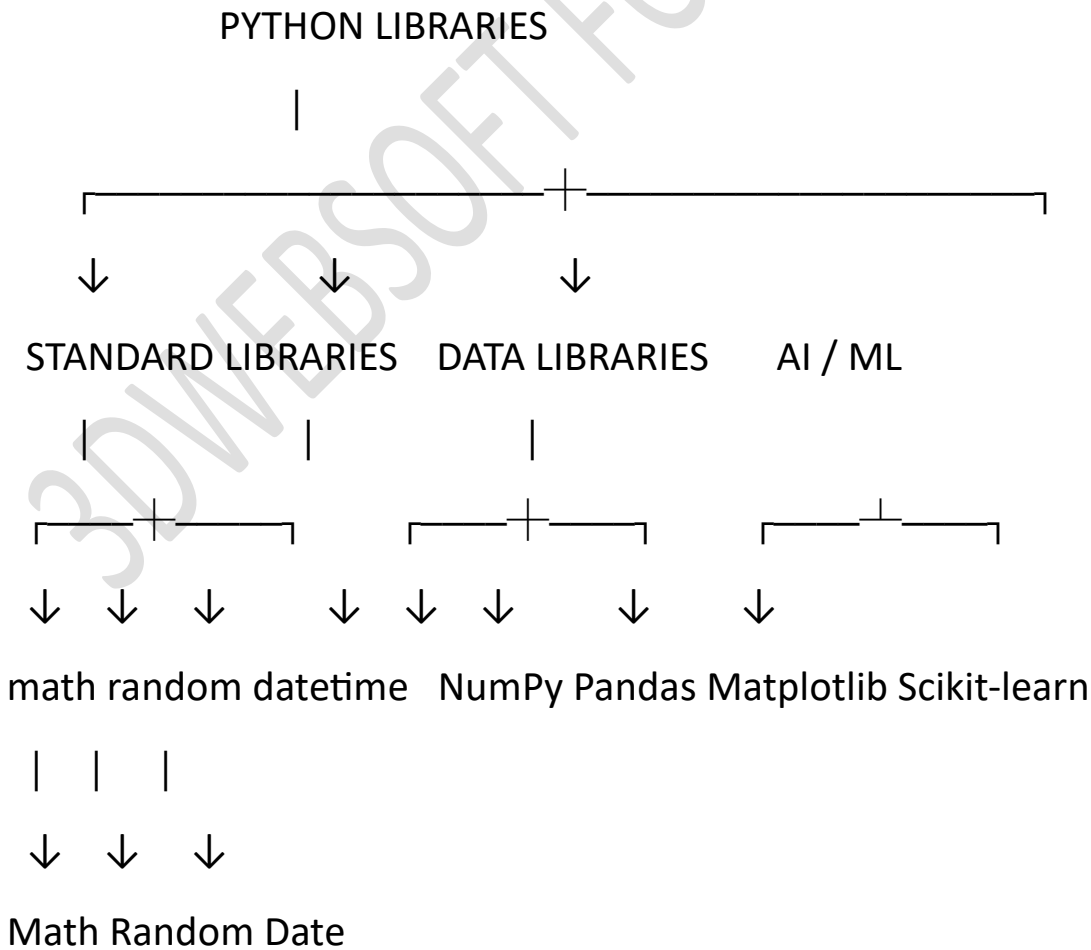
Add:

NumPy

Matplotlib

and create a **Student Performance Dashboard**.

50. Concept Map



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Final Takeaway

Python libraries allow programmers to **reuse powerful, ready-made functionality instead of writing everything from scratch**.

Students should remember:

Library

↓

Ready-made code

↓

Import

↓

Use functions/classes

↓

Build applications

The most important libraries to remember from this class are:

math → Mathematical operations

random → Random values

datetime → Date and time

os → File/system operations

json → JSON data

statistics → Statistics

NumPy → Numerical computing

Pandas → Data analysis

Matplotlib → Charts and graphs

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Requests → APIs/web requests

Scikit-learn → Machine Learning

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